NAME
ChE 210 Quiz #4 27 November 2019

Carbon monoxide (CO) is to be burned with excess air in an adiabatic reactor. Assume 100% conversion of the CO and a basis of 1 mol/s of CO fed to the reactor. The reactants are fed at 25°C and the outlet gases exit at 1400°C. (a) Calculate the percentage excess air fed to the reactor. (b) How would increasing the percentage excess air affect the outlet temperature?

$Q = \emptyset$

25°C

$CO = 1$ mol/sec
EXCESS AIR

AIR
72% n_{1,N_2}
21% n_{1,O_2}

@ 1400 °C

$CO = \emptyset$
$O_2 = n_{2,O_2}$
$N_2 = n_{1,N_2}$
$CO_2 = n_{2,CO_2}$

$2CO + O_2 \rightarrow 2CO_2$

Temp ↑

$PV = nRT$ ↑
↑

SPECIES BALANCE

CO

$_{CO} = n_{1,CO} - 2\beta_1 \rightarrow \beta_1 = \frac{1}{2}$ mol

O_2

$n_{2,O_2} = n_{1,O_2} - \beta_1$

N_2

$n_{2,N_2} = n_{1,N_2}$ → Inert

CO_2

$n_{2,CO_2} = \cancel{n_{1,CO_2}} + 2\beta_1 \rightarrow n_{2,CO_2} =$ mol/sec

BALANCE

$\sum_e n_e \hat{H}_e \quad \sum_i n_i \hat{H}_i + \beta \hat{H}_r$

Table B.8 Specific Enthalpies of Selected Gases: SI Units

25°C @1 ?

$2CO + O_2 \rightarrow 2CO_2$

$\hat{H}$(kJ/mol)

Reference state: Gas, P_{ref} = 1 atm, T_{ref} = 25°C

T	Air	O_2	N_2	H_2	CO	CO_2	H_2O
0	−0.72	−0.73	−0.73	−0.72	−0.73	−0.92	−0.84
25	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	2.19	2.24	2.19	2.16	2.19	2.90	2.54
200	5.15	5.31	5.13	5.06	5.16	7.08	6.01
300	8.17	8.47	8.12	7.96	8.17	11.58	9.57
400	11.24	11.72	11.15	10.89	11.25	16.35	13.23
500	14.37	15.03	14.24	13.83	14.38	21.34	17.01
600	17.55	18.41	17.39	16.81	17.57	26.53	20.91
700	20.80	21.86	20.59	19.81	20.82	31.88	24.92
800	24.10	25.35	23.86	22.85	24.13	37.36	29.05
900	27.46	28.89	27.19	25.93	27.49	42.94	33.32
1000	30.86	32.47	30.56	29.04	30.91	48.60	37.69
1100	34.31	36.07	33.99	32.19	34.37	54.33	42.18
1200	37.81	39.70	37.46	35.39	37.87	60.14	46.78
1300	41.34	43.38	40.97	38.62	41.40	65.98	51.47
1400	44.89	47.07	44.51	41.90	44.95	71.89	56.25
1500	48.45	50.77	48.06	45.22	48.51	77.84	61.09

$\Delta H = \sum_e n_e \hat{H}_e \quad \sum_i n_i \hat{H} + \xi \hat{H}_r^\circ$

$\hat{H}_R$

$2 H_{f\,CO_2} - 2 H_{f\,CO} \quad \emptyset =$

$2 \hat{H}_{CO_2} - [2 \hat{H}_{CO} + \hat{H}_{O_2}] + \hat{H}_R^\circ$

$2(-393.5\ \text{kJ/mol}) - 2(-110.53)$

$2(71.89) - \hat{H}_{O_2} - 565.94 = 0$

$\hat{H}_R = -565.94\ \text{kJ}$

$\hat{H}_{O_2} = -422.16\ \text{kJ of } O_2 \times \frac{1\ \text{mol rxn}}{-565.94\ \text{kJ}}$

$n_{O_2} = 0.75$ mols

$n_{2,O_2} = n_{1,O_2} - \xi_1$

$n_{1,O_2} = 1.25$ mol/sec

$\frac{0.75\ \text{mol}}{1.25} = 0.6 \times 100 = ?$ EXCESS AIR

The tem ature would

a ince $\hat{H}$ is dir to No!